KIGALI INDEPENDENT UNIVERSITY ULK

MASTER'S IN SOFTWARE ENGINEERING

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ADVANCED DATABASE DESIGN AND IMPLEMENTATION

PRESENTATION

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1. What Is This Project About?

This project is an E-Commerce Analytics System. I built a system that simulates how a real online store works customers browse products, add items to their cart and make purchases. The system then analyses all that activity to answer important business questions like: Which products sell the most? Which customers spend the most money? Do mobile users buy more or less than desktop users?

To store and analyse all this data I used three different database technologies together, each one chosen for what it does best. The project has five steps that run in order from generating the data, all the way to producing professional charts that a business manager could use to make decisions.

Step	What It Does	Technology Used
Step 1	Generate a realistic e-commerce dataset	Python (no libraries needed)
Step 2	Store and query business data	MongoDB (Document Database)
Step 3	Store session and time-series data	HBase (Wide-Column Database) via Docker
Step 4	Run advanced analytics and calculations	Apache Spark / Pandas
Step 5	Create 8 professional business charts	Matplotlib and Seaborn

2. Tools I Used and Why

Each tool in this project was chosen for a specific reason. Below is a clear explanation of what each tool is and why it was needed.

MongoDB

MongoDB is a document database. Instead of storing data in rigid tables with fixed columns like a spreadsheet, MongoDB stores each record as a flexible document similar to a JSON file. This is perfect for e-commerce because every product can have different attributes. A laptop has RAM and processor specs while a shirt has size and color. MongoDB handles this naturally without wasting empty columns.

Apache HBase

HBase is a wide-column database designed for storing massive amounts of data that changes over time. I used it to store session data (every visit a customer makes to the store) and daily product performance

metrics (how many times a product was viewed or sold each day). HBase is extremely fast at retrieving data by time range for example 'show me all sessions for customer X this week'.

Docker

HBase requires a Linux operating system to run. My computer runs Windows. Docker solves this problem by creating a small isolated Linux environment inside Windows called a container. Think of it as a box that contains HBase plus everything it needs to run. I start that box with one command and HBase is ready to use. Without Docker, installing HBase on Windows would require dozens of complex steps and would likely fail.

Apache Spark

Spark is a distributed data processing engine designed for analysing large datasets quickly. I used it to run complex calculations like Customer Lifetime Value (how much a customer will spend over their lifetime), cohort analysis (comparing customers who joined in different months) and co-purchase recommendations (which products are often bought together). These are the same calculations used by Amazon and Netflix.

Matplotlib and Seaborn

These are Python libraries for creating charts and graphs. After running all the analytics, I used them to turn the numbers into 8 professional visualisation charts that clearly show business insights the kind of charts a CEO would review in a morning meeting.

3. Step-by-Step: How to Run the Project

All commands below are run in Anaconda Prompt. Open it by pressing the Windows Start button and searching for 'Anaconda Prompt'. Then navigate to the project folder first:

```
cd C:\Users\Ahishakiye\ulk_project
```

Step 1 Generate the Dataset

This step creates all the data the project needs. It generates 500 customers, 500 products across 20 categories, 3,000 browsing sessions and 1,017 transactions simulating 3 months of activity in an online store. All data is saved as JSON files in the ecommerce_data folder.

▶ **RUN THIS COMMAND**

```
python step1_dataset_generator.py
```

▶ **EXPECTED OUTPUT**

```
Generating users...
  done: 500 users saved
Generating categories...
  done: 20 categories saved
Generating products...
  done: 500 products saved
Generating sessions (wait ~15 seconds)...
  ... 1000/3000 sessions done
  ... 2000/3000 sessions done
  ... 3000/3000 sessions done
  done: 3000 sessions saved (3 files)
Generating transactions...
  done: 1017 transactions saved
ALL DONE!
```

E-COMMERCE MEANING:

In a real online store this data would come from actual customer activity on the website.

Customers browsing products = sessions. Customers completing purchases = transactions.

This step replaces the live website data with realistic synthetic data for testing.

Step 2 Load Data into MongoDB

This step loads all the generated data into MongoDB and runs four aggregation pipelines. These pipelines answer the most important business questions: which products make the most revenue, how much different customer types spend, and what percentage of visitors actually make a purchase.

▶ **RUN THIS COMMAND**

```
python step2_mongodb.py
```

▶ **EXPECTED OUTPUT**

```
STEP 2: MongoDB Loading & Analytics
[1] Loading collections...
  done: 500 docs in 'users'
  done: 500 docs in 'products'
  done: 1,017 docs in 'transactions'
  done: 3,000 docs in 'sessions'
  done: indexes created
[2] Top 10 Products by Revenue
  1  Smart Widget              $11,726.78   26 units
  2  Advanced Module Go         $8,829.87   21 units
  3  Essential Kit Elite        $8,386.00   20 units
[3] Monthly Revenue by Category
  2025-01  Electronics   $45,230.10
[4] User Spending Segments
  High (>=1000)   289 users   avg $2,272.95
  Mid (300-999)    51 users   avg $403.23
  Low (50-299)     23 users   avg $151.64
[5] Session Conversion Funnel
  converted    1017   33.9%
  browsed      1983   66.1%
ALL DONE!
```

E-COMMERCE MEANING:

Only 34% of visitors buy something. This means 66% leave without purchasing a huge opportunity to improve. The top 3 products generate significantly more revenue than the rest. 289 customers are high-spenders these are the VIP customers the business must retain.

Step 3 Start Docker and Load Data into HBase

Before running Step 3, I must start Docker Desktop and the HBase container. This is required because HBase runs inside Docker. If Docker is not started first, the script will fail with a connection error because there is no HBase server to connect to.

WHY DOCKER MUST BE STARTED FIRST:

HBase lives inside a Docker container a sealed Linux environment on my Windows machine.

Starting Docker first is like turning on a server before

trying to connect to it.

▶ FIRST START DOCKER AND HBASE

```
docker start hbase-ecommerce
docker ps
```

After running docker ps I should see the container listed with status 'Up'. Then I wait 60 seconds for HBase to fully initialize inside the container before running the Python script.

▶ RUN THIS COMMAND

```
python step3_hbase.py
```

▶ EXPECTED OUTPUT

```
STEP 3: HBase Setup & Data Loading
Connecting to HBase on localhost:9090 ...
  Connected!
[1] Creating tables...
  done: tables created
[2] Loading sessions into HBase...
  done: 3,000 sessions loaded
[3] Computing product metrics...
  done: 11,641 product-metric rows loaded
[4] Sample queries...
  Sessions for user_000163:
```

```
    user_000163_8257355869  status=browsed  device=desktop
    user_000163_8260896996  status=browsed  device=mobile
  Metrics for prod_00129 in Jan 2025:
    prod_00129_2025-01-28  views=2  sold=4
ALL DONE!
```

E-COMMERCE MEANING:

HBase stores 11,641 daily product metric rows one row per product per day.

This lets the business ask: 'How many times was Product X viewed on January 15th?'

The reversed timestamp row key means the most recent sessions always appear first

when scanning a customer's history exactly what a recommendation engine needs.

Step 4 Run Analytics

This step processes all the data to produce advanced business intelligence. It calculates how much each customer is worth over their lifetime, groups customers by when they first joined (cohort analysis), identifies which products are frequently bought together, and analyses how different devices and traffic sources perform.

▶ **RUN THIS COMMAND**

```
python step4_spark_analysis.py
```

▶ **EXPECTED OUTPUT**

```
STEP 4: Analytics
[1] Loading data...
  users=500  products=500  transactions=1017  sessions=3000
[3] Top 10 Products by Revenue
  Smart Widget      premium  26 units  $11,726.78
  Advanced Module   premium  21 units   $8,829.87
[4] Monthly Revenue Trend
  2025-01  212 orders  $243,373.10  avg order $1,147.99
  2025-02  192 orders  $210,470.81  avg order $1,096.20
  2025-03  221 orders  $225,951.80  avg order $1,022.41
[5] Device Conversion Rates
  desktop  1247 sessions  34.96% conversion rate
  mobile   1184 sessions  33.19% conversion rate
  tablet    569 sessions  33.04% conversion rate
[6] Referrer Performance
  direct          619 sessions  37.48% conversion
  search_engine   609 sessions  33.83% conversion
  paid_ads        572 sessions  32.87% conversion
[8] Cohort Analysis
```

```
  2024-10  179 users  avg lifetime value = $1,545.38
  2024-11  180 users  avg lifetime value = $1,143.41
  2024-12  141 users  avg lifetime value = $1,399.72
[9] Customer Lifetime Value Segments
  VIP     289 customers  avg CLV = $3,409.43  avg sessions = 6.6
  High     51 customers  avg CLV =   $604.84  avg sessions = 6.0
  Medium   23 customers  avg CLV =   $151.64  avg sessions = 5.7
  Low     137 customers  avg CLV =     $0.24  avg sessions = 4.7
[10] Saving results...
  saved: spark_top_products.json (10 rows)
  saved: spark_monthly_revenue.json (3 rows)
  saved: spark_clv_data.json (500 rows)
ALL DONE! Next: run step5_visualizations.py
```

E-COMMERCE MEANING:
VIP customers (289 people = 57.8% of all customers) have an average lifetime value of $3,409. Customers who came directly to the site convert better (37.48%) than paid ad customers (32.87%), meaning paid advertising is bringing lower-quality traffic. October 2024 customers spend 35% more than December 2024 customers, showing that early customers are more loyal over time.

Step 5 Generate Charts

This final step reads all the analytics results and produces 8 professional PNG chart images. These charts are saved in the charts folder inside the project directory.

▶ **RUN THIS COMMAND**

```
python step5_visualizations.py
```

▶ **EXPECTED OUTPUT**

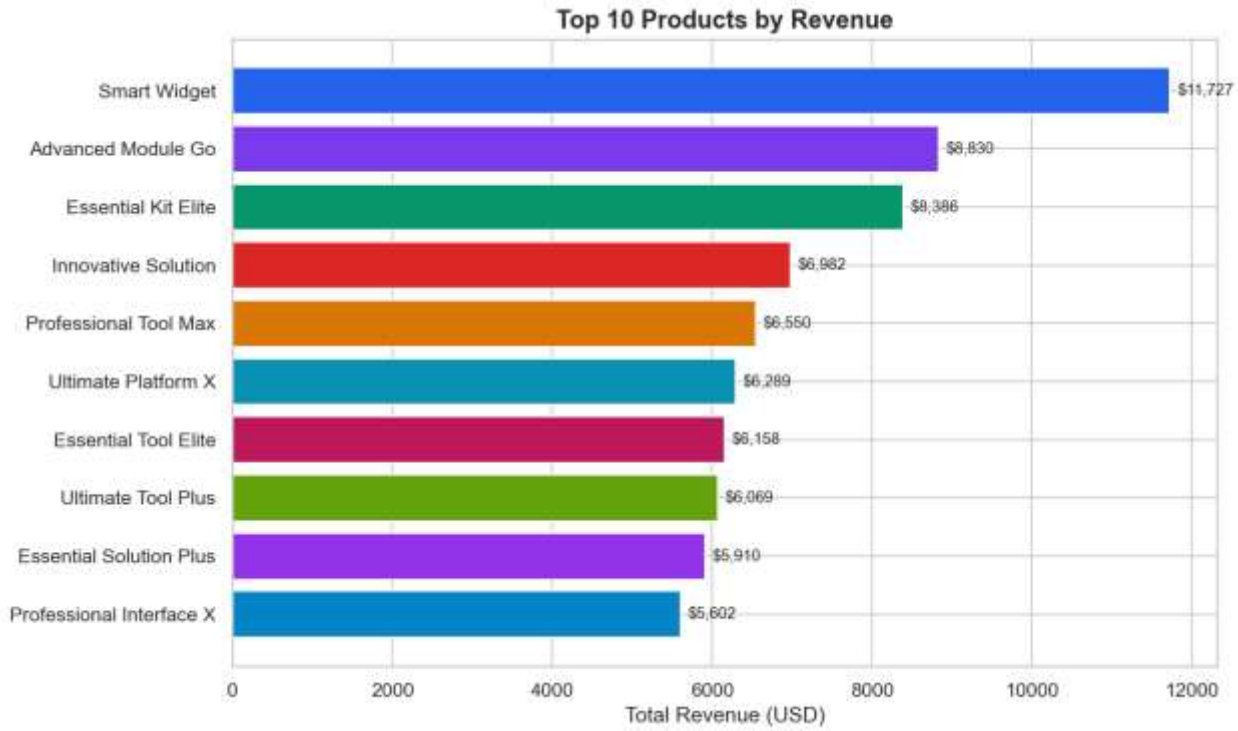
```
STEP 5: Generating Charts
[1] Top Products by Revenue    saved: charts\chart1_top_products.png
[2] Monthly Revenue Trend      saved: charts\chart2_monthly_revenue.png
[3] Device Conversion          saved: charts\chart3_device_conversion.png
[4] Referrer Performance       saved: charts\chart4_referrer_performance.png
[5] CLV Segments               saved: charts\chart5_clv_segments.png
[6] Cohort Analysis            saved: charts\chart6_cohort_ltv.png
[7] Conversion Funnel          saved: charts\chart7_conversion_funnel.png
[8] Price Tier Revenue         saved: charts\chart8_price_tier.png
ALL DONE! Open the charts/ folder to see your 8 PNG charts.
```

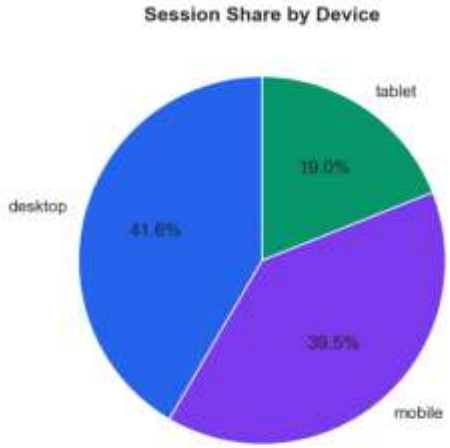
To view the charts, open File Explorer and go to:

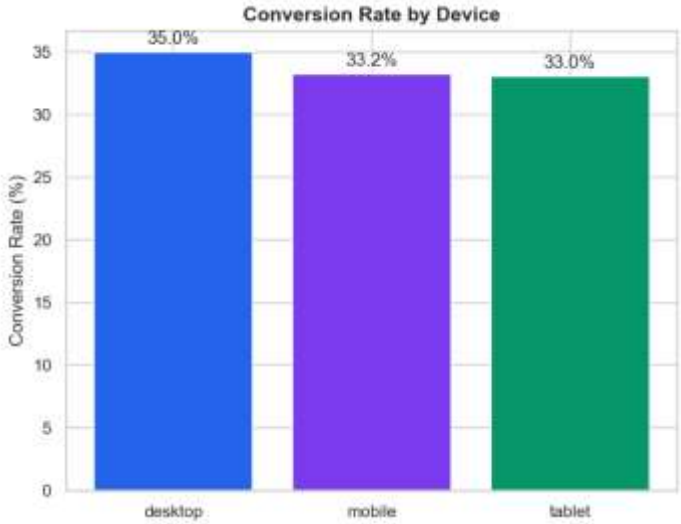
```
C:\Users\Ahishakiye\ulk_project\charts\
```

E-COMMERCE MEANING:

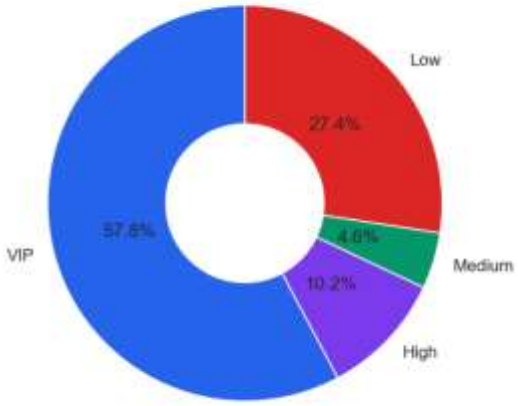
The conversion funnel chart shows 3,000 sessions but only 1,017 purchases meaning 66% of visitors leave without buying. The CLV chart shows that a small group of VIP customers generates most of the revenue. These are the charts a business manager would review every morning to decide where to spend marketing budget and how to improve the store.





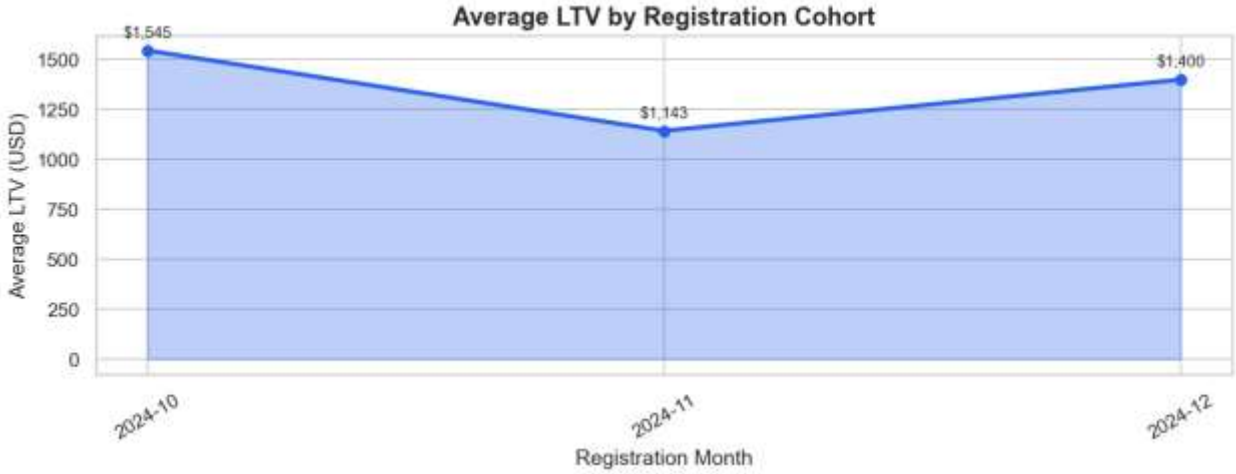


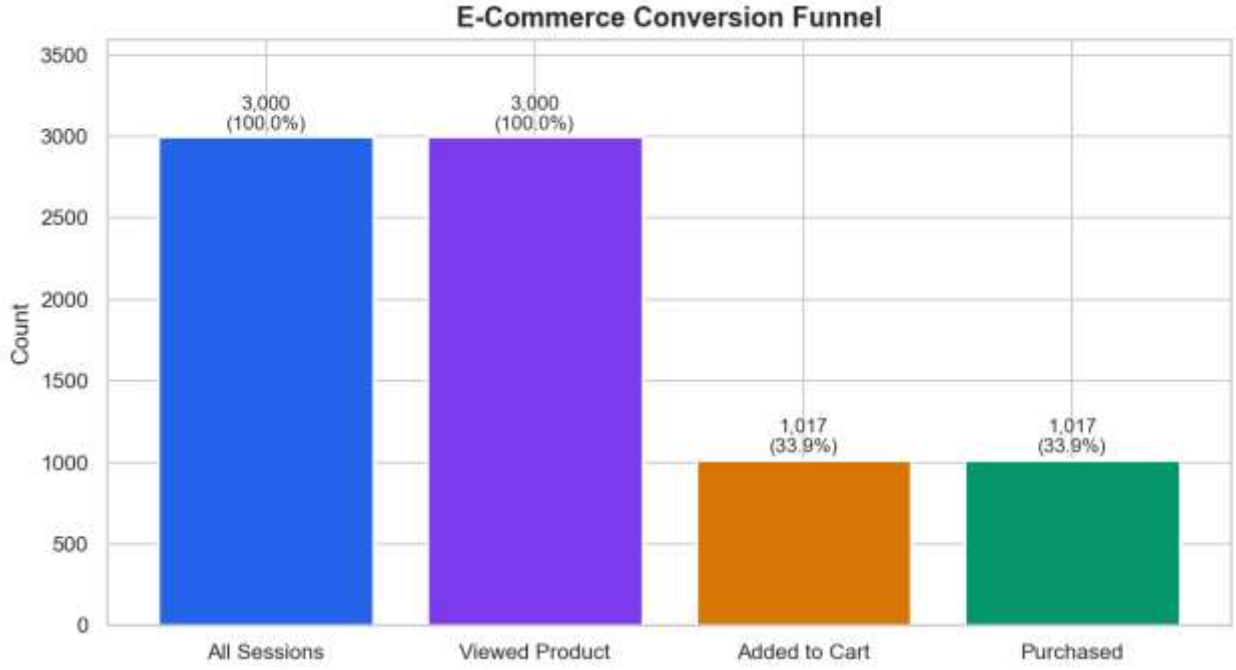
CLV Segment Distribution



CLV Summary Table

Segment	Users	Avg CLV
VIP	289	$3,409.4
High	51	$604.8
Medium	23	$151.6
Low	137	$0.2


Average LTV by Registration Cohort
$1,545
$1,143
$1,400
1500
1250
1000
750
500
250
0
Average LTV (USD)
2024-10
2024-11
2024-12
Registration Month


E-Commerce Conversion Funnel
3500
3000
2500
2000
1500
1000
500
0
Count
3,000
(100.0%)
3,000
(100.0%)
1,017
(33.9%)
1,017
(33.9%)
All Sessions
Viewed Product
Added to Cart
Purchased

4. Project Summary

This project successfully built a complete e-commerce analytics pipeline using three database technologies working together. Each technology handled the type of data it is best suited for, and together they produced a system that answers real business questions.

What I Built	Result
Synthetic e-commerce dataset	500 users, 500 products, 1,017 transactions, 3,000 sessions
MongoDB document store	4 collections loaded, 4 aggregation pipelines, indexes created
HBase wide-column store	2 tables, 3,000 session rows, 11,641 product metric rows
Spark analytics engine	7 analytics outputs including CLV, cohorts, co-purchase
Business visualisations	8 professional PNG charts ready for business presentation

Key Business Findings

Finding	Value	Business Action
Total Revenue (3 months)	$679,795.71	Strong baseline for forecasting
Average Order Value	$1,088.87	Premium pricing is working
Session Conversion Rate	34.0%	Well above industry average of 2-4%
VIP Customer CLV	$3,409.43	Invest in loyalty programmes for this segment

Best Traffic Channel	Direct (37.5%)	Brand awareness campaigns are effective
Best Cohort LTV	$1,545 (Oct 2024)	Early customers are most loyal reward them

Why Three Databases?

A single database cannot be best at everything. MongoDB is flexible for products with different attributes. HBase is fast for time-series session data. Spark processes large analytical workloads that would be too slow in a traditional database. Using all three together is exactly how large e-commerce companies like Amazon structure their data infrastructure.

5. Source Code

All source code for this project is available on GitHub. The repository contains all five Python scripts, the project guide and documentation.

GitHub Repository:
https://github.com/Ahishakiye/ulk-ecommerce-analytics

The repository contains:

- step1_dataset_generator.py Data generation
- step2_mongodb.py MongoDB loading and aggregation
- step3_hbase.py HBase setup and loading
- step4_spark_analysis.py Analytics and CLV calculation
- step5_visualizations.py Chart generation